## Your grade: 98%

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Next item

**1.** A startup is building an AI tool for architects. The tool should take a brief text description — like *"a modern glass office building with a rooftopgarden"* — and produce a completely new visual design that has never existed before. Which fundamental capability of AI makes this possible? 1 / 1 point

- Discriminative AI, which classifies the description into a known architectural style.
- Generative AI, which learns from patterns in existing designs and creates a brand new visual output from the text description.
- Supervised learning, which matches the description to the closest existing design in a database.
- Reinforcement learning, which rewards the system each time it selects the correct design template.

Nice work

Well done! This is a classic application of generative AI. By learning patterns from large amounts of existing architectural imagery and design data, a generative model can take a text prompt and synthesize a completely original visual that has never existed before — turning a description into a new creation.

**2.** A nonprofit organization provides mental health support and wants to make resources available in 12 different languages. They do not have the budget to hire translators for each language. They implement a generative AI tool that translates and adapts their support materials into each language while maintaining appropriate tone and cultural sensitivity. Which statement best explains the real-world significance of generative AI in this situation? 1 / 1 point

- It is significant because it detects errors in the original materials before translation begins.
- It is significant because it makes high-quality, culturally adapted content accessible at a scale that would be financially impossible with human translators alone.
- It is significant because it stores translated documents securely for future use.
- It is significant because it classifies the mental health materials by topic for easier navigation.

Nice work

Well done! This scenario powerfully illustrates the real-world significance of generative AI — it democratizes access to high-quality content by making it possible to generate culturally sensitive, adapted materials across many languages at a fraction of the cost of traditional translation. Generative AI creates new content (the translated materials) rather than simply retrieving or classifying existing ones, delivering meaningful real-world impact.

**3.** A tech company has two AI projects running simultaneously. Project A uses AI to screen job applications and flag candidates as "suitable" or "not suitable" based on their resumes. Project B uses AI to automatically write personalized cover letter templates for job seekers based on their experience. A new team member asks: *"Are both of these generative AI?"* What is the correct response? 1 / 1 point

- No. Project A is discriminative AI because it classifies resumes into categories, and Project B is generative AI because it creates new, original written content.
- Yes, both are generative AI because they both process text as input.
- Both are discriminative AI because both projects are designed to help with hiring.
- No. Project A is generative AI because it makes a decision, and Project B is discriminative AI because it uses existing experience data.

Nice work

Exactly right! This is the core distinction between the two model types. Discriminative AI learns to draw boundaries between categories and classify existing data — exactly what Project A does when it labels resumes. Generative AI learns patterns and produces new, original content — exactly what Project B does when it writes personalized cover letters. Same domain, two fundamentally different AI approaches.

**4.** An EdTech company is enhancing its online learning platform with two new AI capabilities. Capability 1 will assess a student's written essay and assign it a grade from A to F based on quality criteria. Capability 2 will generate a personalized study plan for each student, written as a week-by-week action guide tailored to their learning gaps. The product manager must select the correct AI approach for each capability before development begins. Which selection is correct? 1 / 1 point

- Capability 1 needs a generative model because it reads and understands student writing.
- Capability 2 needs a discriminative model because it uses the student's existing learning data to make decisions.
- Capability 1 needs a discriminative model to classify essay quality into grade categories, and Capability 2 needs a generative model to create personalized, original study plans.
- Both capabilities need discriminative models because both involve evaluating student performance.

Nice work

**Feedback:** Correct! Capability 1 is a clear classification task — the model evaluates existing student work and assigns it to one of several predefined grade categories. Discriminative AI is built for exactly this. Capability 2 requires the creation of something new — a personalized, week-by-week study plan that did not exist before. Generative AI is built for exactly this. Choosing the right model type based on the nature of the output is fundamental to effective AI system design.

**5.** A retail company is deciding between two approaches to flag suspicious return requests. Approach A requires an analyst to write explicit rules, such as 'flag any return over $500 with no receipt.' Approach B is given years of past return records and learns on its own which patterns tend to indicate fraud. Which statement correctly distinguishes these two approaches? 1 / 1 point

- Both approaches are Machine Learning, because both process return data.
- Approach A is Machine Learning because it uses explicit rules, and Approach B is traditional programming because it uses historical data.
- Approach A is traditional programming because a human writes the rules directly; Approach B is Machine Learning because it learns the rules itself from data.
- Both approaches are traditional programming, because both are designed to catch fraud

Nice work

Correct! This is the core distinction: traditional programming relies on human-written rules, while Machine Learning learns patterns directly from data without being explicitly programmed.

**6.** An agricultural technology company is comparing two models for detecting crop disease from field photos. Model A is trained on data, but agronomists must manually specify which visual traits it uses.. Model B is a deep neural network that is fed raw field photos directly and learns on its own which visual traits matter. A team member says: 'Model B is a completely different type of AI than Model A — they aren't related at all.' Is this statement accurate? 1 / 1 point

- Yes, because Model B works with images and Model A works with manually defined traits, so they belong to unrelated fields.
- No, Model B is a Deep Learning model, which is a specialized subset of the same broader Machine Learning field that Model A belongs to.
- Yes, because Model A was developed long before Model B, so they must be part of different fields
- No, both models are traditional programming, since both were designed by the same engineering team.

Nice work

The statement is notaccurate. Deep Learning is not a separate field from Machine Learning — it is a specialized subset distinguished by its use of multi-layered neural networks that learn features automatically, rather than requiring them to be manually defined.

**7.** A university professor asks students to identify correct statements about neural networks in Deep Learning for an end-of-module assessment. **Select all statements that are correct.** 1 / 1 point

- [x] The more layers a neural network has, the deeper it is, which generally enables it to learn more complex patterns from data.

Nice work

Correct! The term "deep" in Deep Learning directly refers to the depth of the neural network — the number of layers it contains. Each additional layer allows the network to learn more abstract and complex representations of the input data. A shallow network might only recognize basic patterns, while a deep network with many layers can recognize highly complex ones — like distinguishing between thousands of different spoken words or identifying objects in images with high accuracy.

- [ ] Neural networks in Deep Learning eliminate the need for any training data because they learn through pre-programmed rules.
- [x] Deep Learning neural networks are a specialized type of Machine Learning model.

Nice work

Correct! Deep Learning is not a separate field — it sits within the broader Machine Learning family. All Deep Learning models are Machine Learning models, but not all Machine Learning models are Deep Learning models. Deep Learning is distinguished specifically by its use of multi-layered neural networks for automatic feature learning. Understanding this hierarchical relationship — AI contains ML, ML contains DL — is fundamental to navigating the AI landscape.

- [x] Neural networks learn by adjusting connection weights between nodes through repeated exposure to training examples

Nice work

Correct! Learning in a neural network happens through a process of repeatedly exposing the network to training examples and adjusting the strength of connections — called weights — between nodes based on how wrong the network's predictions are. Over many training cycles, these weight adjustments accumulate into a network that has effectively "learned" the patterns in the data. This weight adjustment process, guided by backpropagation, is the core mechanism by which Deep Learning models become capable of complex tasks.

- [x] Deep Learning neural networks consist of an input layer, one or more hidden layers, and an output layer, where each layer transforms the data passed to it.

Nice work

Correct! This accurately describes the architecture of a Deep Learning neural network. The input layer receives raw data, the hidden layers progressively transform and learn from that data by extracting increasingly complex patterns, and the output layer produces the final result — whether that is a classification, a prediction, or generated content. The presence of multiple hidden layers is what makes these networks "deep" and capable of learning complex representations.

**8.** A team of AI engineers is onboarding new graduates and explaining how AI has evolved from its earliest forms to modern architectures. **Select all statements that correctly identify characteristics of the transition from rule-based to learning-based AI systems.** 0.8 / 1 point

- [x] One limitation of rule-based systems that motivated the shift to learning-based AI was their inability to handle ambiguous, complex, or unanticipated real-world scenarios.

Nice work

Correct! Rule-based systems worked well within carefully defined boundaries but struggled significantly when faced with real-world complexity. If a situation arose that the rules did not cover — or if inputs were ambiguous or noisy — the system had no mechanism to adapt. This brittleness was a major motivation for developing learning-based approaches, which can handle variability and uncertainty by having learned from diverse real-world examples rather than relying on rigid, pre-defined rules.

- [x] The rise of neural networks represented a shift toward AI systems that could learn representations of data directly, rather than relying on human-engineered features and rules.

This should not be selected

Correct! The rise of neural networks marked a profound architectural shift. Instead of humans defining what features and rules the AI should use, neural networks learn their own representations directly from raw data. This shift — from human-engineered knowledge to machine-learned knowledge — is what enabled AI to scale to problems of far greater complexity than rule-based systems could tackle, and ultimately led to the breakthroughs in image recognition, natural language processing, and generative AI that define modern AI development.

- [x] Early expert systems were a form of rule-based AI that encoded human knowledge as a set of if-then rules to solve problems in specific domains.

Nice work

Correct! Expert systems were one of the most prominent examples of rule-based AI and dominated AI development in the 1970s and 1980s. They worked by encoding the knowledge of human experts — doctors, lawyers, engineers — as a structured set of if-then rules. For example, a medical expert system might contain rules like "if the patient has symptom X and symptom Y, then consider diagnosis Z." While powerful within narrow domains, these systems required enormous human effort to build and maintain, and could not generalize beyond their programmed rules.

- [x] The transition from rule-based to learning-based AI required both algorithmic innovations like backpropagation and increases in available data and computational power.

Nice work

Correct! The transition to learning-based AI was not driven by a single factor — it required a convergence of several developments. Algorithmically, breakthroughs like backpropagation made it possible to train multi-layer neural networks effectively. Practically, the explosion of digital data provided the training material these models needed to learn from. And computationally, advances in hardware — particularly GPUs — provided the processing power to train increasingly large and complex models. All three factors together enabled the shift from rule-based to learning-based AI architectures.

- [ ] Learning-based AI models always outperform rule-based systems in every situation, making rule-based approaches completely obsolete.

**9.** A voice assistant company wants to upgrade their language model so it can hold a natural conversation that references something the user said several minutes and many sentences earlier. Their current RNN-based model frequently' forgets' earlier context. Which architectural change would most directly solve this problem, and why? 1 / 1 point

- Switching to a Convolutional Neural Network (CNN), because CNNs are designed to detect patterns anywhere in an input.
- Switching to a Transformer architecture, because its self-attention mechanism lets the model weigh the relevance of every earlier word directly, rather than passing information step by step.
- Adding more training data to the existing RNN, since more examples will teach it to remember longer conversations
- Increasing the number of decision tree branches in the model, since more branches allow more context to be stored

Nice work

Correct! Self-attention is precisely what solves this problem: instead of relying on information being carried forward step by step, as in an RNN, the model can directly reference any earlier point in the conversation.

**10.** A customer support chatbot built on a standard RNN struggles to reference details a customer mentioned early in a long support conversation, even though it handles short exchanges well. A developer proposes replacing the RNN with an LSTM network. Which statement best explains why this change would help? 1 / 1 point

- LSTMs process the entire conversation in parallel using self-attention, eliminating the need for memory entirely.
- LSTMs introduce gates that control what information is kept or discarded at each step, which helps the network retain important details over longer sequences than a standard RNN can manage.
- LSTMs remove the need for training data by relying on pre-programmed conversation rules.
- LSTMs work by converting the conversation into a decision tree that branches at each customer message.

Nice work

Correct! This is the defining feature of an LSTM: selective gates allow it to carry forward relevant information from much earlier in a sequence, addressing the exact forgetting problem a standard RNN runs into.